Tiago Ferrão-Lindqvist

Data Engineering | Test & Validation Engineer (Mechanical)

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PROFESSIONAL SUMMARY

Mechanical engineer with 12 years of experience in product design, testing and validation across automotive and industrial components. Builds Python tools to process test-rig data, validate results against specifications and generate reports, reducing report preparation from about three days to half a day per test. Uses SQL Server for trend analysis and MATLAB/Simulink for engineering models. Interested in moving into data engineering.

TECHNICAL SKILLS

Python (pandas, NumPy, Matplotlib) | SQL / SQL Server | MATLAB/Simulink | Excel/VBA | Power BI | LabVIEW | Minitab | ANSYS Mechanical | SolidWorks | CATIA V5 | Design of Experiments (DOE) | GD&T | DFMEA | 8D problem solving

PROFESSIONAL EXPERIENCE

Test & Validation Engineer | Corvane Automotive Components — Seat structures and seat-adjuster mechanisms
Setúbal | Mar 2019–Present

- Plan and run durability, vibration and climatic tests across six servo-hydraulic and electric test rigs; prepare test reports for OEM customers.
- Developed Python scripts using pandas, NumPy and Matplotlib to read rig output files, check each channel against the test specification and generate standard report figures. Reduced report preparation time from about three days to half a day per test.
- Query the plant's test-results database in SQL Server for trend analysis across three product generations.
- Correlate rig results with FEA in ANSYS Mechanical and lead 8D root-cause analysis on field failures. Proposed a redesigned adjuster rail bracket that reduced warranty returns for that part by 40%.
- Maintain MATLAB/Simulink models of the adjuster drive, used to select rig load profiles.

Mechanical Design Engineer | Halvik Pumps — Industrial centrifugal pumps
Porto | Sep 2014–Feb 2019

- Designed pump casings and impellers in SolidWorks and produced GD&T production drawings; owned DFMEA for two product lines.
- Conducted structural FEA of casings in ANSYS. Supported the change of one casing family from grey cast iron to ductile iron, saving 12% in material cost.
- Supported the conversion of the pump performance test bench to a PLC-controlled setup using Siemens S7.

EDUCATION

MSc Mechanical Engineering | Sado Valley Polytechnic | 2014
Thesis: Fatigue life prediction of welded brackets under variable-amplitude loading

LANGUAGES

Portuguese (native) | English (fluent) | Spanish (good)